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INTRODUCTION TO NUMBERS

1. **Number :** A number is a mathematical object used to count, measure and label.

2. **Number system** :A numeral system (or system of numeration) is a writing system for expressing numbers, that is, a mathematical notation for representing numbers of a given set, using digits or other symbols in a consistent manner.

3. **International system**:The international system is followed by most of the countries of the world. In this system, a number is split up into groups or periods.In the Inter-national System of Numeration, we use ones, tens, hundreds, thousands, millions and billions.

4. **Decimal system**: Decimal number system, also called Hindu-Arabic, or Arabic Number system, in mathematics, positional numeral system employing 10 as the base and requiring 10 different numerals, the digits 0, 1, 2, 3, 4, 5, 6, 7, 8, 9.

5. **Face value of a number**: Face value is the actual value of the digit. In the number 456, the "5" has a face value of 5.

6. **Place Value of a number** :In our decimal number system, the value of a digit depends on its place, or position, in the number. Each place has a value of 10 times the place to its right.

   Eg : Place value of numneral 5 in the number 584 is 500.

7. **Natural numbers** :Natural numbers, also called counting numbers, are the numbers used for counting things.The set of natural numbers is denoted by N. Where N = {1, 2, 3.....}

8. **Whole numbers**:Whole numbers are positive numbers including zero, without any decimal or fractional parts.The set of whole numbers is denoted by W. Where W = {0, 1, 2, 3.....}

9. **Integers** :Integers are the set of whole numbers and their opposites. The set of integers is denoted by Z .

10. **Positive integer**:Whole numbers greater than zero are called

    positive integers.Positive integers are denoted by Z^+ or N.

11. **Negative integer** :Whole numbers less than zero are called

    negative integers.Negative integers are denoted by Z^-

12. **Note:** The integer zero is neither positive nor negative, and has no sign.

13. **Odd numbers :** An odd number is an integer which is not a multiple of two. If it is divided by two the result is a fraction.

14. **Even numbers**: Even numbers are numbers that can be divided evenly by 2.

15. **Prime number :** A natural number (p > 1) is called a prime number if it is not exactly divisible by any number except '1' and itself.

16. **Composite**: Any natural number which is neither a unit nor a prime is called composite.

17. **Note** : ' 1' is unit . It is neither prime nor composite .

18. **Closure property** :A set of numbers is said to be closed for a specific mathematical operation if the result obtained when an operation is performed on any two numbers in the set, is itself a member of the set. If a set of numbers is closed for a particular operation then it is said to possess the closure property for that operation.

19. **Associative property**: The associative property states that you can add or multiply regardless of how the numbers are grouped.

    By 'grouped' we mean 'how you use parenthesis'. In other words, if you are adding or multiplying it does not matter where you put the parenthesis.

ASSIGNMENT

1. In the International system, the value of 10 lakhs is

   1) 10 Million 2) 100 Thousands

   3) 1 Million 4) 100 Million

2. The numerical 90000000 is equal to

   1) 90 crores 2) 9000 lakhs 3) 9 lakhs 4) 9 crores

3. The smallest 6 digit number having four different digits is

   1) 100000 2) 100023 3) 132001 4) 310002

4. The number of 3-digit numbers in the base-10 system is

   1) 900 2) 999 3) 909 4) 908

5. The place value of 7 in the numeral 27,14, 23, 865 is

   1) Seventy crores 2) Seven crores 3) Ten lakhs 4) Seven

6. The difference of the largest 4-digit number and the smallest 4-digit number is

   1) 1000 2) 9999 3) 9989 4) 8999

7. The sum of the place values of the two 4's in 1400042 is

   1) 400000 2) 399960 3) 39960 4) 400040

8. Numeral for ' sixty million and sixty six ' is _____

   1) 6000066 2)60000066 3)60000606 4) 60000666

9. The difference of the number of 4 digit numbers and the number of 3 digit numbers in base 10 system is

   1) 9900 2) 9000 3) 8100 4) 8000

10. The only true statement in the following is

    1) The number of 4 digit numbers is 9900.

    2) The face value of a digit does not depend on its position in the numeral.

    3) The place value of a digit does not depend on its position in the numeral.

    4) All the above statements are true.

11. The sum of place values of 8 and 5 in 18015 is

    1) 8005 2) 80005 3) 805 4) 85

12. If x is a whole number and x + 3 = 2, then x =

    1) 0 2) 1 3) 2 4) Does not exist

13. The difference between place value and face value of 8 in 18015 is

    1) 7992 2) 8002 3) 7982 4) 8012

14. If a and b are the whole numbers, then ‘ a + b ’ is ____

    1) a natural number 2) Not a whole number

    3) a whole number 4) Not an integer

15. The whole number which does not have a predecessor is

    1) 100 2) 0 3) 1 4) 9

16. 42 × (4 + 2) = (42 × 4) + (42 × 2) is an example of

    1) Closure property 2) Associative property

    3) Identity property 4) Distributive property

SELF TEST

1. Integer which is neither negative nor positive is

   1) 0 2) 1 3) 3 4) 4

2. The smallest 5 digit number having four different digits is

   1) 10000 2) 10023 3) 13201 4) 31002

3. The multiplicative inverse of $3\frac{1}{2}$ is

   1) $\frac{3}{7}$ 2) $\frac{2}{7}$ 3) $\frac{1}{7}$ 4) $\frac{2}{3}$

4. Which of the following is false?

   1) Every rational number is not an integer.

   2) Every surd is not irrational.

   3) Every irrational number is not surd. 4) Every irrational is real.

5. How many rational numbers can be inserted between $\frac{1}{2}$ and $\frac{1}{3}$

   1) 5 2) Infinite 3) finite 4) 6

6. Which of the following number system does not follow the closure property with respect to addition.

   1) Real number system 2) Rational number system

   3) Irrational number system 4) Natural number system

7. The square of a prime number is a

   1) prime number 2) composite number 3) perfect number 4) can't say

8. The number of prime numbers that can be written as a sum and difference of two primes is

   1) 1 2) 2 3) 5 4) 0

LCM AND HCF

1. **Multiple** : A multiple of a number is the product of that number and an integer.
(or)

The integer ‘ a’ is a factor of the integer ‘ b’, or ‘ a ’ divides ‘ b’ if there is an integer q , satisfying b = aq.

If ‘ a’ divides ‘ b’, then ‘ a ’ is called factor of ‘ b’ and ‘ b ’ is called multiple of ‘ a’.

Notation: ‘ a ’ divides ‘ b’ is denoted by a/b. The notation for

“ a does not divide b ” is a $\nmid$ b.

2. **Prime Factor** : A factor of a given number is called a prime factor, if it is a prime number.

3. **Common factor** :A number which divides each one of the given numbers exactly, is called a common factor of the given numbers.

4. **Modulus function: (Absolute Value Function)**. Asolute Value Function f(x) = |x| is defined as x for nonnegative values of x and -x for negative values of x.

5. **Relatively prime (Coprime)**:Two integers ‘a’ and ‘ b’ are said to be relatively prime, mutually prime, or coprime (also spelled co-prime) if the only positive integer that evenly divides both of them is 1.

6. **Dividend = Divisor x Quotient + Remainder .**

7. **LCM**

(i) LCM of Natural Numbers:

The least common multiple (LCM) of two or more **natural numbers** is the least of their common multiples.

SIMPLIFICATION

(BODMAS Rule, addition and subtraction of fractions, simplification of polynomials, properties of indices, finding square roots.)

BODMAS RULE

Easy and simple way to remember BODMAS rule!!

B $\rightarrow$ Brackets first (parentheses)

O $\rightarrow$ Of (orders i.e. Powers and Square Roots, Cube Roots, etc.)

DM $\rightarrow$ Division and Multiplication (start from left to right)

AS $\rightarrow$ Addition and Subtraction (start from left to right)

Note:

(i) Start Divide/Multiply from left side to right side since they perform equally.

(ii) Start Add/Subtract from left side to right side since they perform equally.

Fraction : A fraction (from Latin: fractus means "broken") represents a part of a whole or, more generally, any number of equal parts.

Simple fraction: A simple fraction (Also known as a **common fraction** or **Vulgar fraction**) is a rational number written as $\frac{a}{b}$, where a and b are both integers ($b \neq 0$).

ASSIGNMENT

1. If a – [2a – {3b – (4c – 2a)}] is simplified, then the answer is

   1) a + 3b + 4c 2) a – 3b – 4c 3) a + 3b – 4c 4) a – 3b + 4c

2. If 3x – 6x + (2y – x) is simplified, then the answer is

   1) 4x – 2y 2) –4x – 2y 3) –4x + 2y 4) 4x + 2y

3. If – p – {2q – (6p + 4q) is simplified, then the answer is

   1) 2q – 5p 2) 2q + 5p 3) –2q – 5p 4) – 2q + 5p

4. If A = $6x^2 - \{2x - (3x^2 - 7x - 3)\}$ and B = $2x^2 - \{4x + (2x^2 - 7x + 6)\}$, then B – { – (A + 3B)} is

   1) $9x^2 - 3x - 27$ 2) $9x^2 + 3x - 27$ 3) $9x^2 + 3x + 27$ 4) $-9x^2 + 3x - 27$

5. p – [p – (q + p) – {p – (2p – q)}] =

   1) – 2q 2) 2q 3) – 2p 4) – 3p

6. If $\frac{a^4b^2}{c^3d^2} = \frac{2^4\times3^2}{3^3\times5^2}$, then the equivalent fraction of $\frac{a\times d}{b\times c}$ is

   1) $\frac{2^7\times5^2}{2^6\times3^2}$ 2) $\frac{2^7\times5^1}{2^6\times3^2}$ 3) $\frac{2^7\times5^2}{2^5\times3^3}$ 4) $\frac{2^7\times5}{2^4\times3^3}$

7. If $\frac{2}{a}+\frac{3}{a}+\frac{5}{a} = 1$, $\frac{7}{b}+\frac{3}{b}+\frac{12}{b}+\frac{8}{b} = 2$ and $\frac{13}{c}+\frac{14}{c}+\frac{15}{c}+\frac{18}{c} = 5$, then the value of $\frac{b+a}{b-a}\times\frac{c-b}{c+b}\times\frac{c+a}{c-a}$ in reduced form is

   1) $\frac{55}{9}$ 2) $-\frac{55}{9}$ 3) $-\frac{54}{9}$ 4) $-\frac{52}{9}$

8. If $\frac{x\times y}{A\times B} = \frac{5\times8}{9\times7}$, $\frac{P\times Q}{R\times S} = \frac{3\times14}{10\times4}$, then $\left(\frac{x\times y}{P\times Q}\right)\div\left(\frac{A\times B}{R\times S}\right)$ in power notation is

   1) $\frac{5^2\times2^4}{3^3\times7^2}$ 2) $\frac{5\times2^5}{3^3\times7^2}$ 3) $\frac{5^2\times2^5}{3^3\times7^2}$ 4) $\frac{5^2\times2^5}{3^2\times7^2}$

9. If $\frac{3}{5}$ of $\frac{2}{3}$ is added to – 1, then the resultant is

   1) $\frac{-3}{5}$ 2) $\frac{3}{5}$ 3) $\frac{4}{5}$ 4) $-\frac{4}{5}$

FACTORIZATION

Factorization:

1. **Definition :** In mathematics, factorization (also factorisation in some forms of British English) or factoring is the decomposition of an object (for example, a number, a polynomial) into a product of other objects, or factors, which when multiplied together give the original.

2. Factoring is mainly used for solving or simplifying polynomial expressions. The process of polynomial factoring is also called as the resolution into factors.

3. There are three methods (common factors, grouping terms and using formulas) are used to find polynomial factors in common .

i. Factoring Polynomials by Common Factor :

First we have to find the common factor for the given polynomial equation. If a given algebraic polynomial equation A contains common factor B, then divide each term of equation A by common factor B and you will get an expression C. Now B and C are the factors of A and it can be expressed as A = B x C

ii.Factoring Polynomials by Grouping terms:

If the Polynomial Equation doesn't have common factors then group the terms. By grouping terms in an appropriate manner you can get a common factor.

RATIO AND PROPORTIONS

Synopsis

RATIO:

1. The ratio of two quantities of the same kind and in the same unit is the fraction that one quantity is of the other.

2. A ratio is simply a fraction. The following notations all express the ratio of a to b ; a : b , $a \div b$, or $\frac{a}{b}$.

3. In the ratio a : b , 'a' is called the first term or antecedent and 'b' is called the second term or consequent.

PROPORTION:

4. A proportion is simply an equality between two ratios (fractions).

5. Two variables are directly proportional if one is a constant multiple of the other i.e., y = kx where 'k' is a constant.

6. If one quantity increases (or decreases) while another quantity decreases (or increases), the quantities are said to be inversely proportional. The statement " y is inversely proportional to x " is written as $y = \frac{k}{x}$ where 'k' is a constant.

7. If two numbers are in the ratio of a:b, then, for some number 'x' ,the first number is 'ax' and the second number is *'bx'.*

8. If a set of objects is divided into two groups in the ratio of a:b, then the first group contains $\frac{a}{a+b}$ of the objects and the second group contains $\frac{b}{a+b}$ of the objects.

9. If a set of objects is divided into three groups in the ratio a:b:c, then the first group contains $\frac{a}{a+b+c}$ of the objects the second $\frac{b}{a+b+c}$, and the third $\frac{c}{a+b+c}$.

10. **Commensurable quantities:** Two quantities are said to be commensurable if their ratio can be expressed as the ratio of two integers. Otherwise they

25. If 40% of a number is equal to two-third of another number, what is the ratio of first number to the second number.

1)2:5 2)2:7 3)5:7 4)5:3

26. The ratio of number of boys and girls is 4 : 3. If there are 18 girls in a class, then the total number of students in the class eqaul to

1)42 2)24 3)16 4)48

27. A hare and a jackal are running a race.Three leaps of the hare are equal to four leaps of the jackal. For every six leaps of the hare,the jackal takes 5 leaps.Ratio of their speeds.

1)64:25 2)8:5 3)5:8 4)25:64

28. The ratio of Science and Arts students in a college is 4:3.If 14 Science students shift to Arts then the ratio becomes 1:1.Total strength of Science and Arts students.

1)156 2)169 3)225 4)196

29. In a class of 70 students, the ratio of boys and girls is 4:3.Among the girls the ratio of the number of rich and poor is 1:4.If the ratio of the number of rich and poor students is 8:27.The ratio of the number of rich and poor boys in the class

1)1:3 2)1:4 3)1:2 4)2:3

30. Rs. 600 was divided among P, Q, R and S so that P and Q got four times as much as R and S together, Q got 3 times of what R got and R got $1\frac{1}{2}$ times as much as S. Then P got Rs.....

1) 216 2)264 3)72 4)none

SELF TEST KEY

1. 2	2. 1	3. 3	4. 4	5. 3	6. 4	7. 3	8. 1	9. 3	10. 1
11. 3	12. 1	13. 4	14. 3	15. 1	16. 2	17. 3	18. 4	19. 3	20. 4
21. 3	22. 3	23. 2	24. 1	25. 4	26. 2	27. 2	28. 4	29. 1	30. 2

WORD PROBLEMS

AGES , AVERAGE ,TIME ,PERCENTAGES ,PROFIT LOSS

PERCENTAGES

1. In mathematics, a percentage is a number or ratio expressed as a fraction of 100. It is often denoted using the percent sign, "%", or the abbreviations "pct.", "pct"; sometimes the abbreviation "pc" is also used. A percentage is a dimensionless number (pure number).

   Thus, x percent means x hundredths, written as x%.

   To express x% as a fraction: We have, x% = $\frac{x}{100}$.

2. **Percentage Increase/Decrease:**

   If the price of a commodity increases by R%, then the reduction in consumption so as not to increase the expenditure is

$$\left[\frac{R}{(100+R)}\times 100\right]\%$$

   If the price of a commodity decreases by R%, then the increase in consumption so as not to increase the expenditure is

$$\left[\frac{R}{(100-R)}\times 100\right]\%$$

3. **Results on Population**: Let the population of a town be P now and suppose it increases at the rate of R% per annum, then:

   1. Population after ' n' years = $P\left(1+\frac{R}{100}\right)^n$.

   2. Population ' n ' years ago = $\frac{P}{\left(1+\frac{R}{100}\right)^n}$